

SUDHAKAR RODDA

Embedded Software Engineer | AUTOSAR | MCAL | Functional Safety | Diagnostics

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SUMMARY

Embedded Software Engineer with 6+ years of experience in AUTOSAR Classic, RTOS-based systems, and embedded firmware development for automotive ECUs.

Expertise in MCAL development, CDD/low-level drivers, ECU bring-up, system integration, and system-level debugging, with strong knowledge of diagnostics (DCM, DEM), ECU mode management (EcuM, BswM), and AUTOSAR COM stack (COM, PduR, CanIf).

Hands-on experience in CAN, UDS diagnostics, and working knowledge of J1939, with strong focus on Functional Safety (ISO 26262), including safety mechanisms, SMU integration, and MISRA-C compliant development on AURIX and S32K platforms.

Basic exposure to AUTOSAR Cybersecurity (SecOC, CSM, Crypto) and Memory Stack (NvM, Fee, Ea), with strong experience in end-to-end ECU development and real-time debugging.

SKILLS

- **Programming:** C, Embedded C for automotive embedded systems.
- **AUTOSAR & Embedded Systems:** AUTOSAR Classic (MCAL + BSW), CDD/low-level driver development, ECU bring-up, system integration, and RTOS concepts (task scheduling, interrupts, resource management).
- **MCAL & BSW Modules:** MCAL drivers (ADC, PWM, GPT, DIO, SPI); BSW modules (COM, PduR, CanIf, DCM, DEM, EcuM, BswM).
- **Communication & Protocols:** CAN (ISO 11898), UDS (ISO 14229); working knowledge of J1939 (TP, NM, DM, PGN/SPN); interfaces: SPI, I2C, UART.
- **Functional Safety:** ISO 26262 (HARA, ASIL, safety mechanisms, SMU) and MISRA-C compliant development.
- **Microcontrollers:** Infineon AURIX (TC3xx), NXP S32K3, Renesas RH850; motor control ICs (TLE9189, TLE9180).
- **Debugging & Tools:** TRACE32, iSystem, WinIDEA; CANoe/CANalyzer; DaVinci, EB Tresos/C4K, RTA-CAR, Git, PTC-ALM.
- **System Expertise:** End-to-end debugging across MCAL, BSW, and application layers with register-level analysis.
- **Cybersecurity & Memory:** AUTOSAR SecOC, CSM, Crypto and Memory Stack (NvM, Fee, Ea) - basic exposure
- **Process & Build:** Automotive SDLC (V-Model, Agile); GCC and GHS toolchains.

EDUCATION

- Bachelor of Technology (B. Tech) – Electronics and Communication Engineering (ECE)
Jawaharlal Nehru Technological University (JNTU), Kakinada | 2019

PROFESSIONAL EXPERIENCE & PROJECT HIGHLIGHTS

Embedded AUTOSAR / Firmware Software Engineer

Embitel Technologies Pvt Ltd (Volkswagen Group Company) Sep 2024 – Present

Project: EV Motor Control – Advik

- Developed AUTOSAR-based embedded firmware for EV motor control (IPMSM) (NXP S32K312).
- Integrated MCAL & BSW (COM, PduR, CanIf) and performed ECU bring-up.
- Developed CDD and low-level drivers using MCAL (PWM, ADC, DMA, SPI, GPIO, Port, GPT, MCU, WDG).
- Implemented SPI/I2C/UART communication and hardware interfacing.
- Implemented RTOS-based tasks, interrupts, and resource management.
- Implemented UDS diagnostics (DCM/DEM) and CAN communication stack
- Performed system integration and end-to-end debugging across MCAL, BSW, and application layers
- Implemented ISO 26262 safety mechanisms, SMU integration, and MISRA-C compliance.

Project: EV Motor Control – Cummins

- Developed AUTOSAR-based firmware for EV motor control ECU (AURIX TC399).
- Implemented MCAL drivers and CDD for motor control peripherals.
- Developed RTOS-based control tasks with real-time scheduling and interrupt handling.
- Integrated SPI/I2C/UART for hardware interfacing and sensor/actuator communication.
- Implemented fault detection, diagnostics (DCM/DEM), and CAN-based error handling.
- Performed system-level debugging and issue analysis using TRACE32
- Implemented Functional Safety (ISO 26262) mechanisms including SMU integration

Embedded Autosar Software Engineer

Veoneer Technologies (Pacecom Technologies Pvt Ltd), Feb 2023 – Sep 2024

Veoneer Technologies (Kpit Technologies Pvt Ltd), May 2022 – Jan 2023

Project: Restraint Control System (RCS) – Mazda

- Developed and integrated AUTOSAR SWCs for safety-critical ECU.
- Implemented diagnostics (DCM/DEM) and improved ECU reliability.
- Performed cross-layer debugging (Application + BSW) with system-level analysis.
- Applied RTOS concepts (task scheduling, interrupts, resource management) in system behavior.
- Contributed to Functional Safety (ISO 26262) including safety concepts and MISRA-C compliance.
- Performed functional and timing (chronometric) testing.
- Exposure to AUTOSAR Cybersecurity (SecOC, CSM, Crypto).

Associated Software Engineer

HYUNDAI, July 2021 - Jan 2022

Project: Advanced Driver Assistance Systems (ADAS)

- Developed AUTOSAR-based ECU features for ADAS systems.
- Configured CAN communication and DCM (DID) diagnostics.
- Performed cross-layer debugging (Application + BSW).
- Implemented and validated diagnostics (DTC, DID) using DCM/DEM.
- Ensured MISRA-C compliance and coding standards.

Trainee Engineer

HYUNDAI (YASHASWI), July 2019 - July 2021

- Supported AUTOSAR and embedded firmware development (analysis, coding, testing)
- Implemented features and performed debugging and issue resolution.
- Followed coding standards and participated in code reviews

DECLARATION

I hereby declare that the above-written particulars are true to the best of my knowledge and belief.

Place: Bengaluru

Name: Sudhakar Rodda